

中国科学院大学
University of Chinese Academy of Sciences

Generative Models & AI4S Applications

Course Overview

Fall 2026

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About the Teaching Team

LAMP

机器学习与感知实验室

- **Instructor**
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  - **Research Group**
    - ✓ Learning And Machine Perception (LAMP) Lab
    - ✓ http://lamp.ucas.ac.cn/
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About this Course

- This course will be taught in Chinese, with **slides in English**

- This is a **newly offered course**, so the arrangements may be adjusted as the semester progresses

- **Announcement:** Course materials are developed with reference to related courses from **leading international universities**
  - MIT, CMU, Cornell, etc.
  - and other online materials

Prerequisites

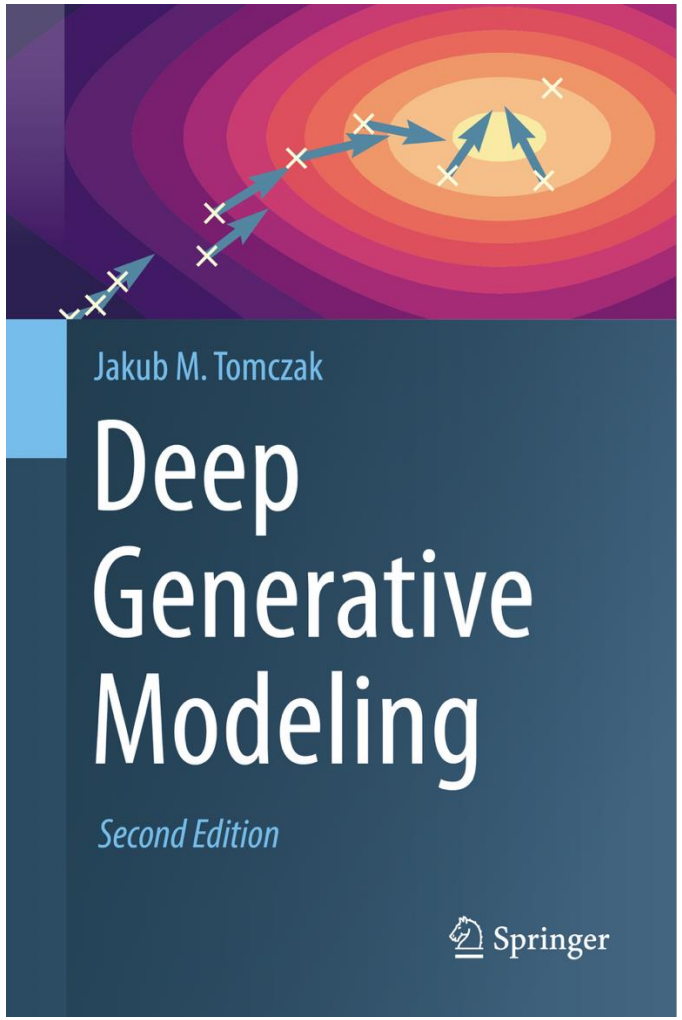
- **Required**
  - **Math:** Calculus, Linear Algebra, **Probability Theory**
  - Machine Learning
  - Deep Learning
- **Recommended**
  - Optimization
  - Mathematical Statistics
  - Python Programming
    - ✓ Practice in PyTorch and Deep Learning Project

Who is this Course FOR / NOT FOR

- **This Course is FOR Students who are…**
  - **Interested** in core principles of generative models
  - **Eager** to study generative model families systematically
  - **Self-motivated** to learn beyond lectures
  - Comfortable with mathematical and technical details
- **This Course is NOT FOR Students who are…**
  - Looking only for hands-on AI tool tutorials
  - Expecting an extremely light survey without math
  - Only interested in prompt engineering but not model principles

Course Materials

- **Textbook**



- **Deep Generative Modeling (2nd Ed.)**
  - **Author:** Jakub M. Tomczak
  - **Publisher:** Springer, 2024
  - **Companion Code:** https://github.com/jmtomczak/intro_dgm
- **Covers various major families of deep generative models**

Course Materials

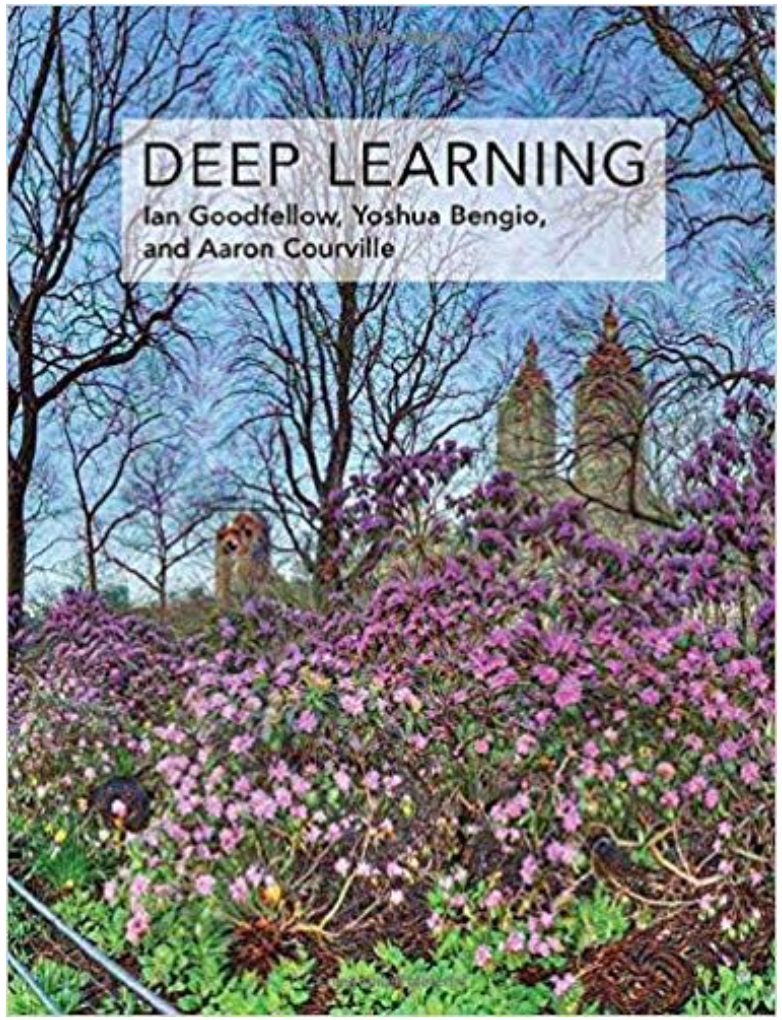
- **Textbook**



- **The Principles of Diffusion Models**
  - **Author:** Chieh-Hsin Lai, Yang Song, Stefano Ermon, et al.
  - **Publisher:** MIT Press, 2027 (forthcoming)
  - **Official Page:** https://the-principles-of-diffusion-models.github.io/
- **Focuses on central topics related to diffusion models**

Course Materials

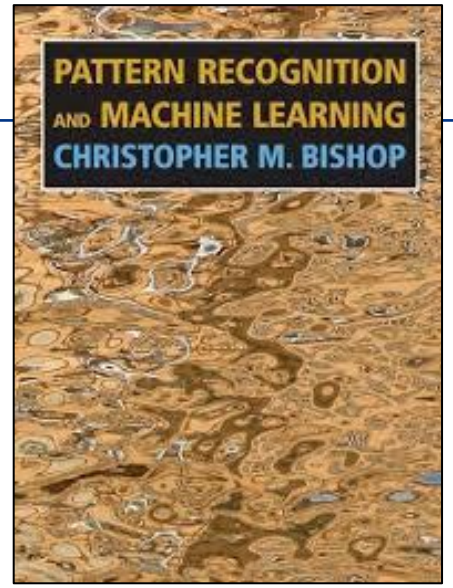
- **Textbook for Deep Learning**

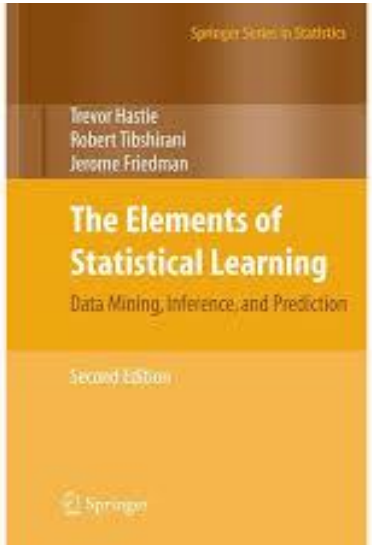


- **Deep Learning**
  - **Author:** Ian Goodfellow, Yoshua Bengio, Aaron Courville
  - **Publisher:** MIT Press, 2016
  - **Companion Code:** https://github.com/janishar/mit-deep-learning-book-pdf
- **There is also a Chinese version**

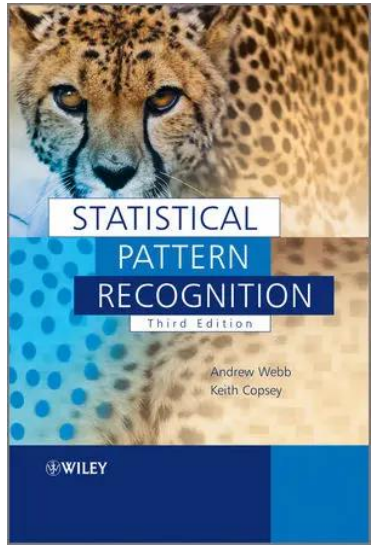
Course Materials

- **Textbook for Machine Learning**
  - Pattern Recognition and Machine Learning (PRML)
  - Christopher M. Bishop
  - https://prml.github.io/

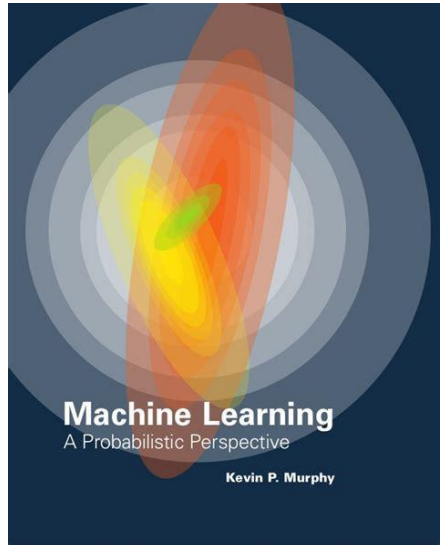




The Elements of Statistical Learning, Springer, 2001



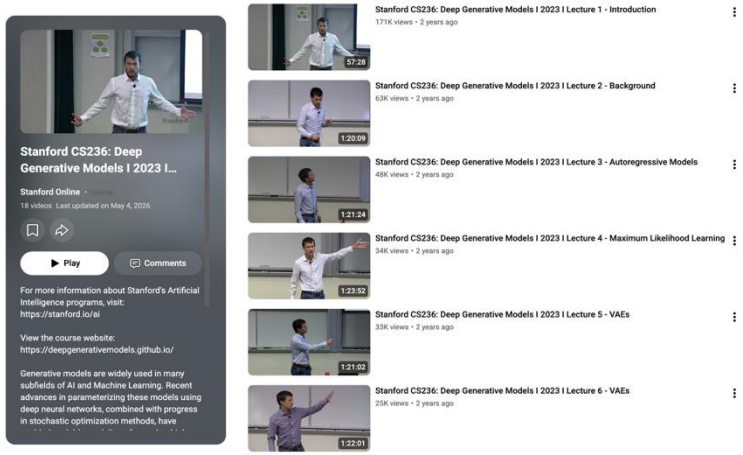
Statistical Pattern Recognition, 3rd Edition, Wiley, 2011

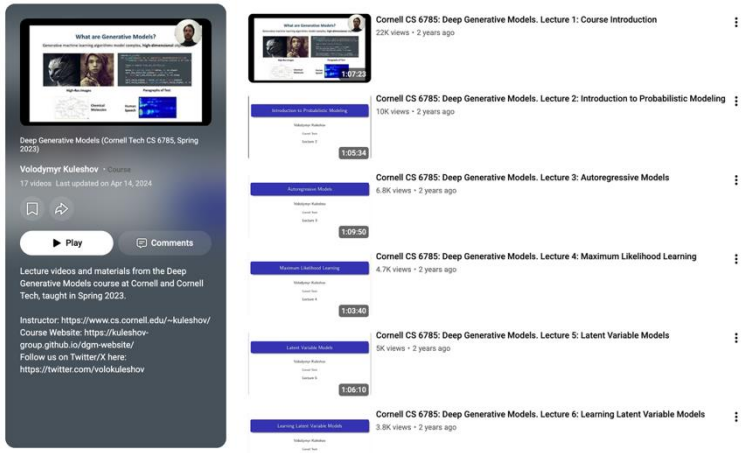


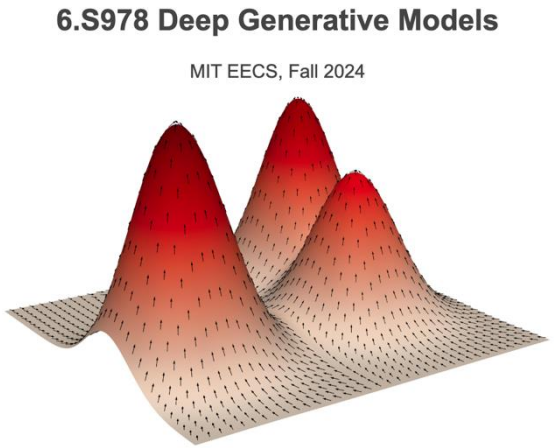
Machine Learning: a Probabilistic Perspective, Murphy, 2012

Course Materials

- **Online Material: Deep Generative Models**







Stanford CS 236

- **Stefano Ermon**
- https://deepgenerativemodels.github.io/
- Slides and Recording

Cornell CS 6785

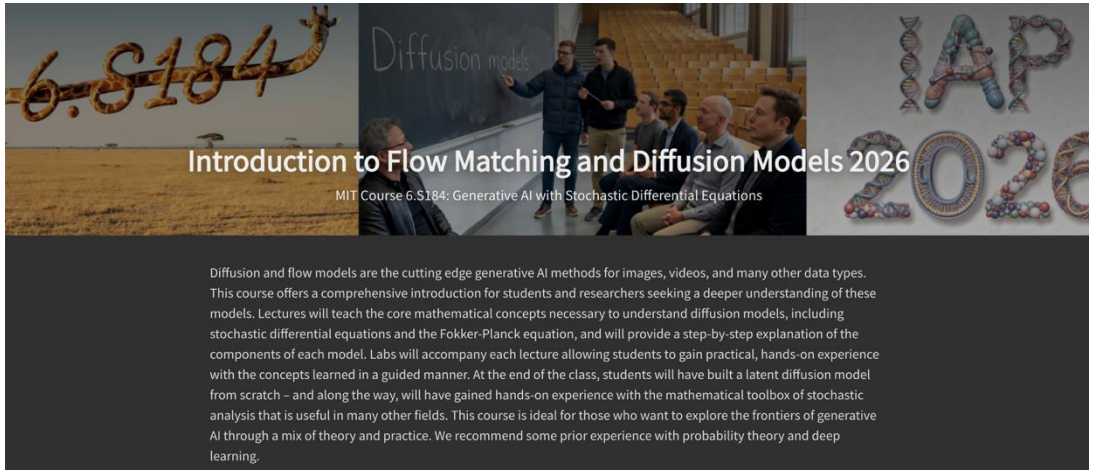
- **Volodymyr Kuleshov**
- https://kuleshov-group.github.io/dgm-website/
- Slides and Recording

MIT 6.S978

- **Kaiming He**
- https://mit-6s978.github.io/
- Slides, Reading Lists and Labs

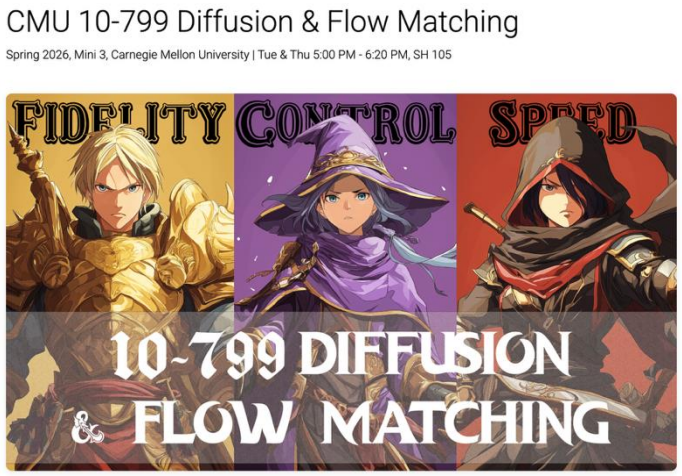
Course Materials

- **Online Material: Diffusion and Flow Matching**



MIT 6.S184

- **Generative AI with SDE / ODE**
- https://diffusion.csail.mit.edu/2026/
- Slides, Recording, Labs, and Notes



CMU 10-799

- **Selected Topics related to Diffusion**
- https://kellyyutonghe.github.io/10799S26/
- Slides, Recording, HWs

Syllabus

- **General Topics**
  - Formulation, Training, and Evaluation
- **Specific Models**
  - Autoregressive (AR) Models
  - Latent Variables Models (VAE)
  - Normalizing Flows (NF)
  - Generative Adversarial Networks (GANs)
  - Diffusion / Score-based / Flow Models
- **Applications in AI4Science**
  - Remains an introductory review

- **Focus on principles and model properties**
- **Not a survey of latest trends or frontiers**

Grading Policy

- **Attendance & Participation (10%)**
- **Assignments (50%)**
  - Release and submit online
  - Conceptual and programming problems
  - GPU recommended, but not required
  - **Assignments are designed to deepen understanding**
- **Final Exam (40%)**
  - Open-book
  - Focus on conceptual understanding



群聊: 26秋-生成模型&AI4S-课程群

该二维码7天内(9月13日前)有效，重新进入将更新